DENTALFLOW

Shared Patient Intelligence Architecture & Implementation Roadmap

Updated direction: Practice Operations + Patient Communication + AI X-ray Analysis

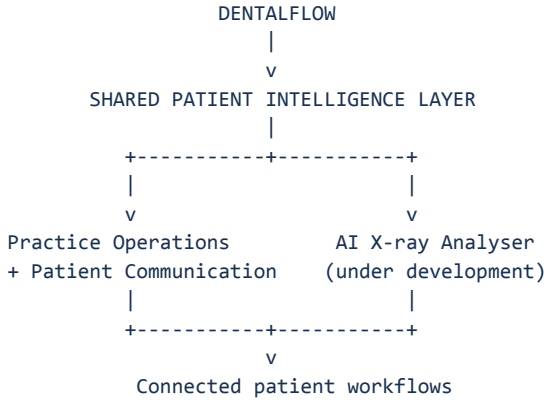
Prepared for: Mohona
Defines the revised product direction, shared patient foundation, system connections and implementation order.

Start with dentistry. Build for healthcare. Scale globally.

1. The Updated Product Direction

DentalFlow is intended to become one connected healthcare ecosystem rather than a collection of separate products. We are starting with dentistry and building from the platform already deployed.

- Practice Operations + Patient Communication — the existing DentalFlow platform.
- AI X-ray Analyser — a planned module currently under development.
- Shared Patient Intelligence Layer — the common foundation that connects both.



Scope decision

The Consultation Analyser has been removed from the roadmap. We will not build an automated module that listens to patient–dentist conversations and generates treatment reports. This does not prevent DentalFlow from storing clinician-entered consultation notes or treatment plans as ordinary patient records.

Why the Consultation Analyser is out of scope

The research and risk review identified material reliability concerns: accent and language variation, background noise, overlapping speech, speech-recognition errors, and the possibility that a generative model could introduce details that were never said. A plausible-looking but incorrect treatment report could damage clinical trust. For these reasons, the Consultation Analyser is removed rather than treated as a near-term development module.

The roadmap remains focused on the shared patient foundation, the existing operations and communication platform, and the AI X-ray Analyser. X-ray outputs must be clearly labelled as potential findings for clinician review—not autonomous diagnoses or treatment decisions.

2. What We Already Have

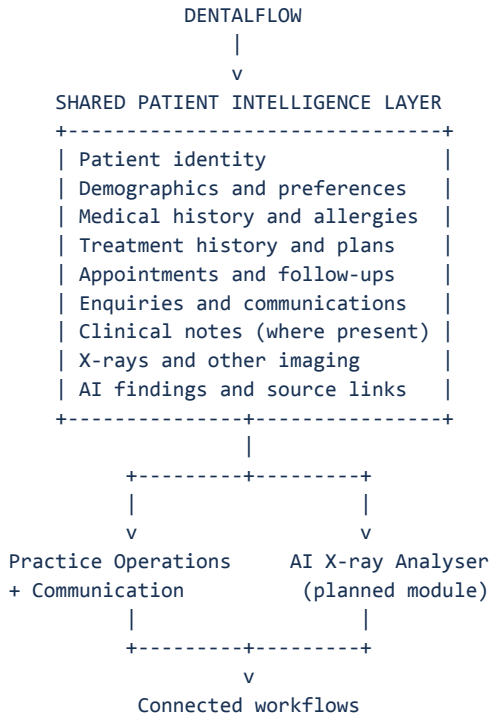
The first part of DentalFlow is built and deployed. It is the starting point for the architecture and should be connected to the shared foundation rather than rebuilt.

- Patient enquiries and communication
- Appointments and scheduling workflows
- Follow-ups and patient reactivation
- Alerts and operational visibility
- Analytics and workflow tracking
- AI-assisted operational features

Implementation principle
Audit and understand the existing product first. Reuse working capabilities and data where appropriate; do not create a parallel platform.

3. Target Architecture

The shared layer is the common patient context for DentalFlow modules. Each module should access it through defined APIs/services, not by depending directly on another module's internal code.



Core architectural idea

Build the patient foundation once. Connect the existing platform to it. Add X-ray intelligence as a separate, replaceable module using the same patient identity and context. Modules should share context through stable interfaces and controlled data access.

4. How the Systems Should Communicate

Use a common API/service layer and, where useful, an event layer to connect modules. Avoid direct dependencies between module internals.

```
Module action
     |
     v
API / Service Layer
     |
     v
Shared Patient Intelligence Layer
     |
     v
Event or permitted action
     |
     v
Relevant DentalFlow module
```

Example: patient enquiry to follow-up

```
Patient enquiry
     |
Identify or match patient
     |
Retrieve permitted patient context
     |
Communication and appointment workflow
     |
Appointment / cancellation / follow-up
     |
Record outcome and update patient timeline
```

Example: X-ray analysis

```
X-ray attached to the correct patient record
     |
Validate file, patient association and permissions
     |
AI X-ray Analyser processes the image
     |
Potential findings + limitations where supported
     |
Store structured output and link to original image
     |
Clinician reviews, accepts, edits or rejects findings
     |
Only clinician-confirmed information informs clinical decisions
     |
Relevant operational action, if authorised
```

Clinical safety boundary

An AI-generated finding must not automatically become a confirmed diagnosis, treatment plan or patient-facing claim. Preserve the original image, label AI output clearly, and require appropriate clinician review before clinical action.

5. The Shared Patient Intelligence Layer

This is the most important architectural component: a longitudinal patient record that authorised DentalFlow modules can use. There should be one canonical patient identity across the platform, with appropriate controls for practice and organisation boundaries.

Patient and identity information

- Unique DentalFlow Patient ID
- Demographics and contact details
- Relevant communication preferences
- Patient matching and duplicate-resolution status

Medical and treatment information

- Medical history and allergies, where collected and authorised
- Previous treatments and treatment history
- Clinician-entered clinical notes and treatment plans, where available

Operational information

- Appointments, cancellations and attendance
- Enquiries and communication history
- Follow-up and reactivation history
- Operational events and relevant outcomes

Imaging and AI information

- Original X-ray or imaging file and metadata
- AI-generated potential findings, model/version and timestamp
- Source references linking findings to image evidence where supported
- Clinician review status and corrections
- Audit trail of relevant changes

One patient identity

The X-ray Analyser must not create a separate patient database or independent patient identity. It should associate each image and analysis with the existing DentalFlow patient record, subject to identity verification and permissions.

6. Step-by-Step Implementation Roadmap

Do not begin by building the X-ray Analyser in isolation. First establish how the existing platform stores and relates patient data, then build the foundation and connect the current product.

Phase 1 — Audit the current DentalFlow platform

Document what already exists before changing the data model.

- Current database and patient tables
- Appointment, enquiry, communication and follow-up records
- Patient files and storage
- Existing APIs and integrations
- Relationships between records
- Authentication, user roles and organisation/practice structure

- Current access controls and audit capabilities

Phase 2 — Design the shared patient model

Define Patient as the canonical entity and document its relationships.

```
Patient
├── Appointments
├── Enquiries
├── Communications
├── Follow-ups
├── Treatments / Treatment Plans (where available)
├── Clinical Notes (where available)
├── X-rays / Imaging
└── AI Analysis and Review History
```

Decide how records are matched, what fields are required, how duplicate records are resolved, and how data is scoped across practices and organisations.

Phase 3 — Build the Shared Patient Intelligence Layer

- Find and identify a patient safely
- Read authorised patient information
- Add or update records with validation
- Retrieve relevant history
- Connect records to the canonical patient ID
- Track record changes and preserve audit history

Phase 4 — Build the API/service foundation

Define versioned, permission-aware APIs for patient identity, appointments, communications, follow-ups, imaging and AI-analysis records. Keep interfaces stable so modules can evolve independently.

Phase 5 — Introduce an event layer where justified

Define business events for important state changes. Choose the implementation after reviewing the current stack and expected scale.

- PatientCreated / PatientUpdated
- AppointmentBooked / AppointmentCancelled
- FollowUpRequired
- XrayUploaded
- XrayAnalysisCompleted
- AIFindingReviewed
- TreatmentPlanUpdated (when clinician-confirmed and supported)

An event should not imply clinical approval. XrayAnalysisCompleted means analysis has finished; it does not mean findings have been confirmed.

Phase 6 — Connect the existing DentalFlow platform

Make Practice Operations + Patient Communication the first major consumer of the shared patient layer. Validate that appointments, communications, enquiries and follow-ups are associated with the correct patients and existing workflows continue to work.

Phase 7 — Build the AI X-ray Analyser

- Define supported image types and quality checks
- Associate images with the correct patient and encounter
- Generate structured potential findings rather than ungrounded free-form reports

- Preserve links to the original image and analysis version
- Make uncertainty and limitations visible where technically supported
- Provide a clinician review workflow
- Record corrections, approvals and rejections
- Keep AI output separate from clinician-confirmed records

Phase 8 — Connect X-ray intelligence to operational workflows

After clinical review, authorised and confirmed information may support appropriate follow-up or appointment workflows. The system should not independently diagnose, prescribe or commit the clinician to a treatment decision.

Phase 9 — Validate and expand

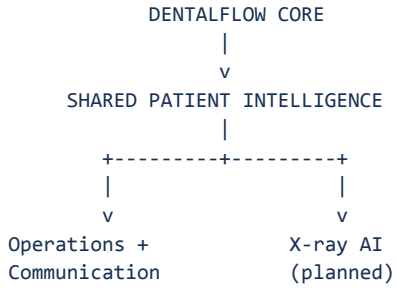
Evaluate the module with qualified dental professionals and representative images before broad deployment. Define acceptance criteria for image quality, performance, false positives, false negatives, workflow usability, auditability and safe failure behaviour. Expand only when validation and applicable regulatory requirements support it.

7. Technical Principles

Principle	What it means
One patient identity	Every module uses the same canonical patient ID and a safe matching process.
One shared data model	Avoid multiple conflicting copies of the same patient information.
API-based communication	Modules communicate through defined, versioned services.
Events where useful	Important state changes can be published for authorised consumers.
Loose coupling	The X-ray module can be upgraded or replaced without breaking the core platform.
Structured AI outputs	Represent findings as validated fields and records, not only narrative text.
Preserve source information	Retain the original image, metadata, analysis version and traceability.
Clinician review and provenance	Distinguish AI-generated potential findings from clinician-confirmed information.
Permissions from the beginning	Enforce organisation/practice isolation, role-based access and auditability.
Privacy and security by design	Use appropriate encryption, secure storage, retention controls and access logging.
Build for expansion	Add future intelligence modules without rebuilding the shared foundation.

8. What Not to Build

- Do not create a separate patient database for the X-ray Analyser.
- Do not create duplicate patient identities for different DentalFlow modules.
- Do not use spreadsheets, CSV exports, manual copy/paste or ad hoc file transfers as the core integration method.
- Do not tightly couple the X-ray module to internal frontend or backend implementation details of the existing platform.
- Do not treat AI-generated potential findings as confirmed diagnoses or automatically convert them into treatment decisions.
- Do not allow unsupported AI-generated details into the patient record without provenance and review.



9. First Development Milestone

The first milestone is not the X-ray model. It is the foundation that makes the current platform and future imaging intelligence work together safely.

First milestone
Shared Patient Intelligence Layer + API foundation + existing DentalFlow connected.

Questions the first sprint must answer

1. What patient data exists today, and where is each field stored?
2. What should the canonical Patient model look like?
3. Which records belong to a patient and how are relationships represented?
4. How do appointments, enquiries, communications and follow-ups connect to a patient?
5. How will imaging files and AI analyses connect to the same patient?
6. Which APIs are needed, and what are their permission boundaries?
7. Which events are needed for important workflow transitions?
8. What roles and access controls apply across practices and organisations?
9. How will current data be migrated or mapped without losing history?
10. How will the system prevent or resolve duplicate patient records?

First concrete deliverable

Produce a concise architecture specification containing:

- Canonical Patient ID and patient schema
- Entity relationship diagram
- API/service list and access rules
- Data migration and duplicate-resolution plan
- Event list and event semantics
- Security, permissions and audit requirements
- Plan to connect existing DentalFlow workflows
- Acceptance tests for data integrity and practice isolation

10. AI X-ray Analyser: Product and Safety Boundaries

The X-ray Analyser is a planned module. It should analyse supported dental images, identify potential findings and produce structured information that a qualified dentist can review. The objective is to support clinical assessment and connect useful information to the patient workflow—not to replace professional judgement.

Expected high-level workflow

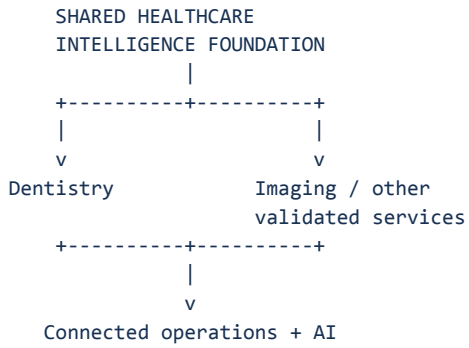
```
X-ray upload
   |
Patient association + permission checks
   |
Image quality / supported-format checks
   |
AI analysis
   |
Structured potential findings + traceability
   |
Dentist reviews and edits / accepts / rejects
   |
Confirmed clinical record, if appropriate
   |
Linked operational workflow, where authorised
```

Safeguards to plan for

- Clearly distinguish AI output from clinician-confirmed findings.
- Do not generate unsupported findings or treatment recommendations.
- Preserve source images and analysis versions for traceability.
- Record reviewer identity, review status and edits where appropriate.
- Communicate uncertainty and handle poor-quality or unsupported images safely.
- Do not claim validated diagnostic accuracy until supported by appropriate evaluation.
- Assess privacy, security, clinical safety and applicable regulatory requirements before deployment.

11. Longer-Term Expansion

Once the architecture works reliably for dentistry, the shared foundation may be extended to additional healthcare domains and specialised intelligence services. This is a long-term direction, not a claim that those modules already exist.



The reusable foundation remains shared identity, shared data, contextual relationships, APIs, events, permissions and auditability. Any future module must be evaluated separately for safety, reliability, clinical validity and commercial fit before entering the roadmap.

12. Final Direction

We are not building disconnected products. We are building one DentalFlow platform with an existing operational capability, a shared patient foundation, and an AI X-ray capability planned to use that same foundation.

```
CURRENT
Practice Operations + Patient Communication
                  |
                  v
SHARED PATIENT INTELLIGENCE LAYER
                  |
                  v
AI X-ray Analyser (under development)
                  |
                  v
Connected clinical context + operational workflows
                  |
                  v
DENTALFLOW ECOSYSTEM
                  |
                  v
Long-term healthcare intelligence expansion
```

Revised development order

Audit existing platform → Shared Patient Model → APIs / Services → Events where useful → Connect Existing DentalFlow → AI X-ray Analyser → Clinician Review & Validation → Connect Approved Outputs to Workflows → Consider Healthcare Expansion

Scope reminder

The Consultation Analyser is not part of the current or planned roadmap in this document. Consultation records and treatment plans may still exist as ordinary patient data where entered or confirmed by authorised clinical staff.

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